

## Coding & Solution

|                      |                                         |
|----------------------|-----------------------------------------|
| <b>Date</b>          | 10 July 2026                            |
| <b>Team ID</b>       | xxxxxx                                  |
| <b>Project Name</b>  | Rising Waters – Flood Prediction System |
| <b>Maximum Marks</b> | 5 Marks                                 |

### Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

#### Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

### Solution Summary

| Field                                | Details                                                                                                                                                                  |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Repository Link / URL</b>         | [Paste your GitHub / GitLab or hosted link here]                                                                                                                         |
| <b>Programming Language(s)</b>       | [e.g., Python, JavaScript, Java]                                                                                                                                         |
| <b>Framework(s) Used</b>             | [e.g., Flask, React, Spring Boot]                                                                                                                                        |
| <b>Key Features Implemented</b>      | Data preprocessing, Flood prediction using Machine Learning, Model training, Prediction system, Data visualization                                                       |
| <b>Pending / Incomplete Features</b> | Real-time weather API integration and live flood alert notification (or None if everything is completed)                                                                 |
| <b>Setup / Run Instructions</b>      | Install Python libraries from requirements.txt, open the Jupyter Notebook/Google Colab, load the dataset, run all cells, and use the trained model for flood prediction. |

### Code Quality Checklist

| S.No | Criteria                                               | Status (Yes / No) |
|------|--------------------------------------------------------|-------------------|
| 1    | Code is modular and organized into functions / classes | Yes               |
| 2    | Meaningful variable and function names are used        | Yes               |
| 3    | Code includes comments / documentation where necessary | Yes               |
| 4    | Error handling is implemented for critical operations  | Yes               |
| 5    | The application runs without critical errors           | Yes               |
| 6    | Code is committed to a version control repository      | Yes               |

### Additional Notes / Comments:

The Flood Prediction System is developed using Python and Machine Learning with a well-structured and modular codebase. Meaningful variable names, proper documentation, and error handling are implemented. The project is maintained in a GitHub repository, making it easy to manage, reuse, and enhance in the future.